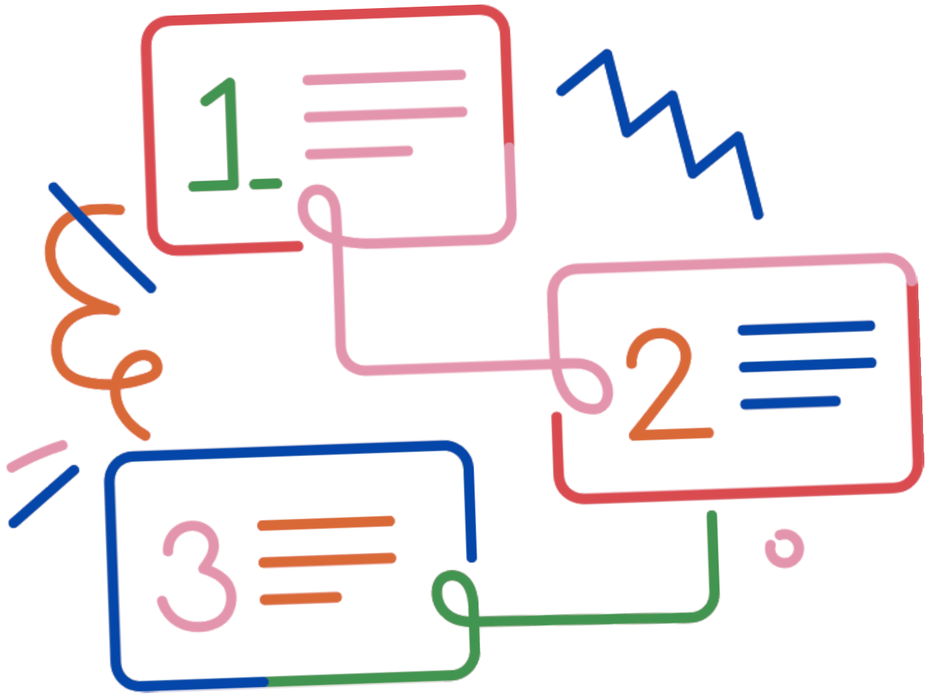
MAI 2026

Phase1-Design Project Requirements



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Introduction

The project aims to provide on-demand virtualized computing resources, including servers, storage, and containers, with automatic scaling to meet workload changes. It will be built on an open-source, modular architecture supporting private and hybrid cloud deployments. The system will feature self-healing infrastructure, efficient orchestration, and a centralized dashboard for resource management, ensuring seamless deployment and optimized resource use.

Objectives

Promote Sustainable Living

Deploy an innovative web application that contributes to achieving one or more of the United Nations Sustainable Development Goals (SDGs) within the establishment.

Efficient Cloud-Based Solution



Deliver a private, scalable, and secure cloud infrastructure to host and manage the application, ensuring high availability and performance.

Multi-Tenant Architecture



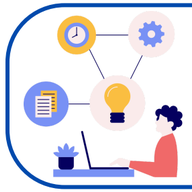
Enable isolated and secure instances for each client or group of users, promoting privacy and customization.

Automation and Orchestration



Simplify deployment and lifecycle management through automation and centralized orchestration, reducing repetitive tasks and provisioning times.

Problems To Solve



Resource Management:

Inefficient allocation of computing and storage resources in traditional infrastructures.



User Engagement:

Lack of personalized and predictive user experiences in existing systems.



Automation Gaps:

Manual provisioning and monitoring processes lead to delays and errors.



Application Scaling:

Inability to handle high traffic or workload fluctuations efficiently



Data Security:

Insufficient mechanisms to ensure privacy and prevent breaches in multi-tenant environments

Expected Outcomes

Sustainable Impact

A web-based application that actively promotes sustainable practices, aligned with one of the SDGs.

Optimized Infrastructure



Scalable, resilient, and secure cloud infrastructure supporting automated scaling and resource allocation.

Enhanced User Experience



A web-based application designed to deliver an intuitive, accessible, and seamless user experience through responsive design, efficient navigation, and personalized interactions.

Simplified Management



Centralized orchestration and monitoring tools for easy deployment, maintenance, and performance visualization.

AI Integration



Modular and portable architecture that supports hybrid cloud deployments and integrates with various AI models for continuous improvement.

Infrastructure Solution (IaaS)

- Open Source
- Multi-tenant / modular architecture
- Native Load Balancer & floating IP
- Centralized dashboard for resource management
- Orchestration capabilities
- Supports private & hybrid deployment
- Scale vertically & horizontally / data redundancy
- Auto-scaling and Self-healing

openstack

apachecloudstack
open source cloud computing

cloudforet

Virtualization Solution

Compatible with Linux, multiple hypervisors

Open-sourceSupports high scalability for demand

Infrastructure Management Compatibility

Open-source

Live migration support for VMs needed

Support for demanding application workloads









Infrastructure Automation Solution

- Open-source
- Smooth infrastructure connection
- Resource relationship specification

- Human-readable templates (IaC)
- Advanced integration & high availability







Containerization

Enables efficient multiple deployments

Installable on Debian/Ubuntu

Supports load balancing and service discovery

Supports declarative config and automation



Orchestration Solution

Linux Distribution Compatibility

Automated installation

Deployment Environment as a Service

Service discovery & load balancing

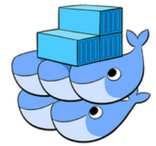
Support for declarative configuration & automation

{

kubernetes

RED HAT OPENSHIFT

DOCKER SWARM



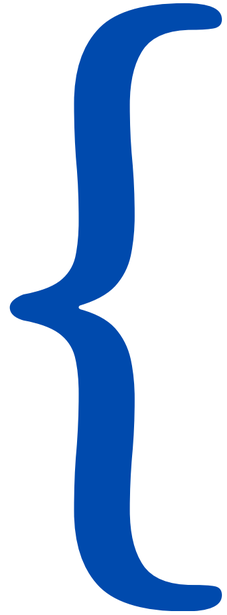
Monitoring & Visualization Solution

Performance monitoring capabilities

Agentless

Visualization & automation of dashboards

Standard protocols





Grafana



Prometheus

ZABBIX

Nagios

Configuration Automation Solution

Open-source

Agentless Operation

Provisioning, configuration, and deployment.

Standardized Protocols (e.g., SSH)

Platform Compatibility (Windows, Linux, Unix)



Technological Choices













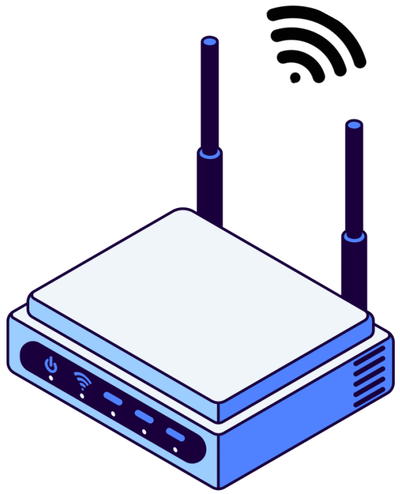


Grafana



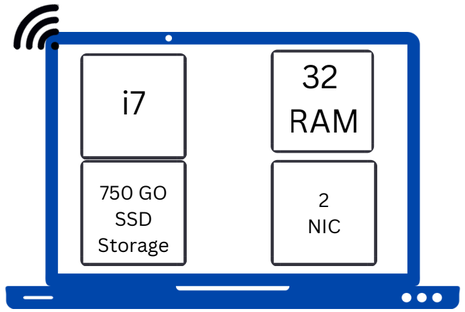
Prometheus

Physical architecture

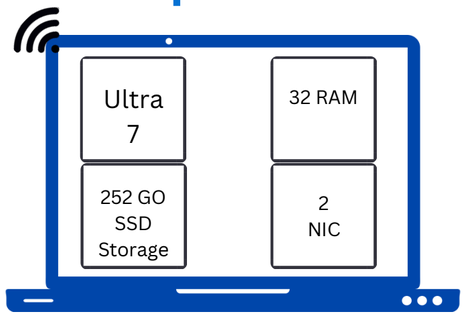


Router

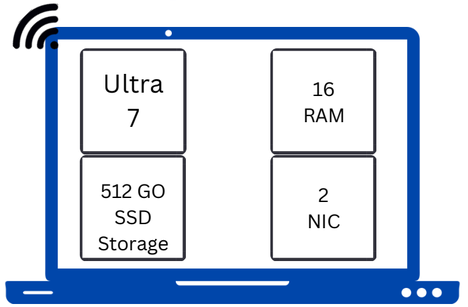
Controller Node



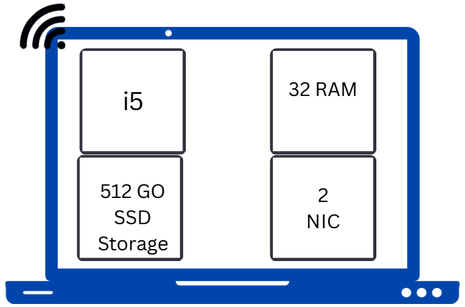
Compute Node

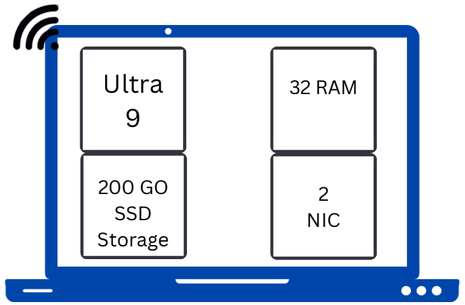


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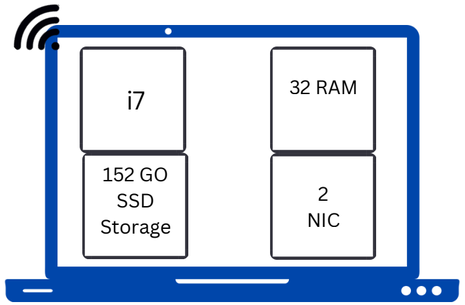


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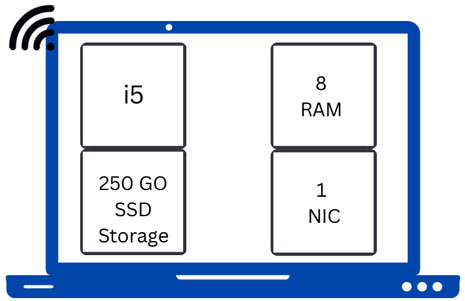




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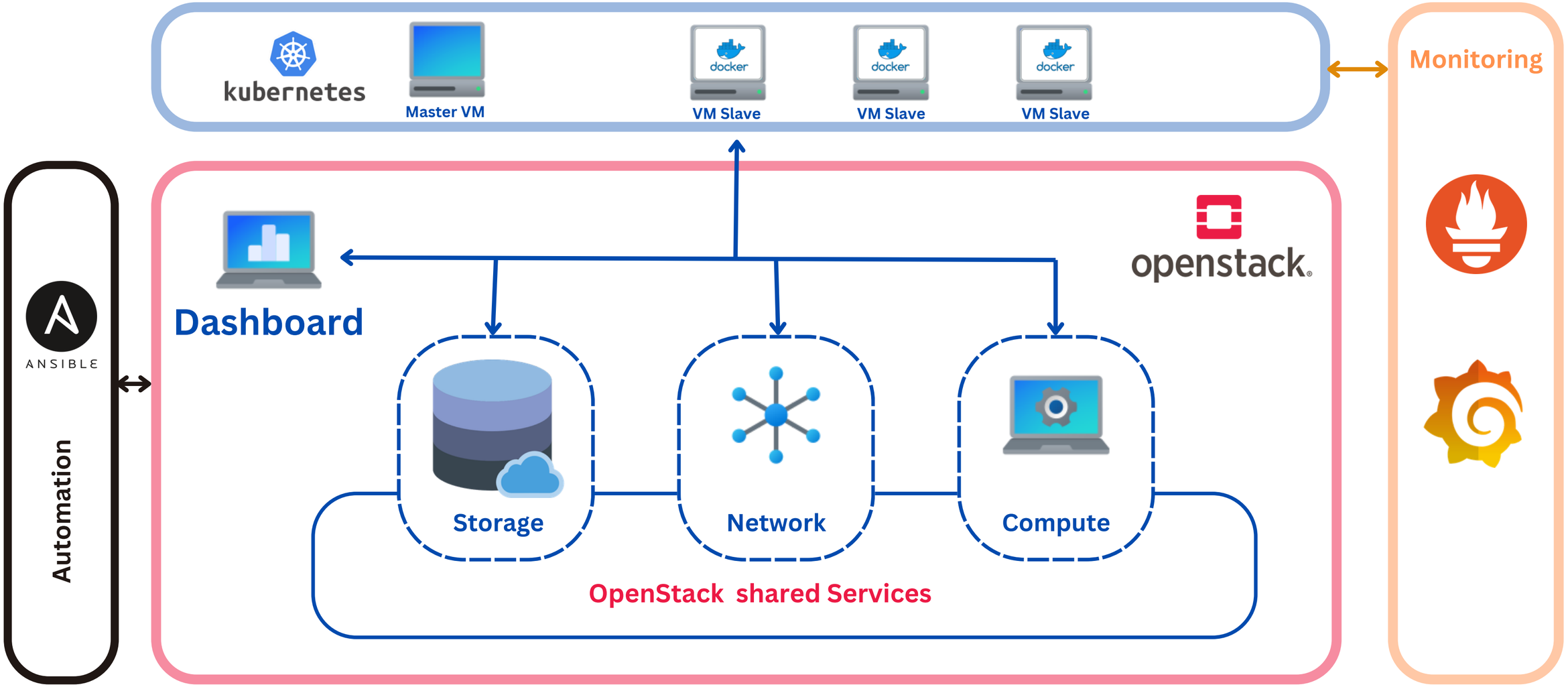


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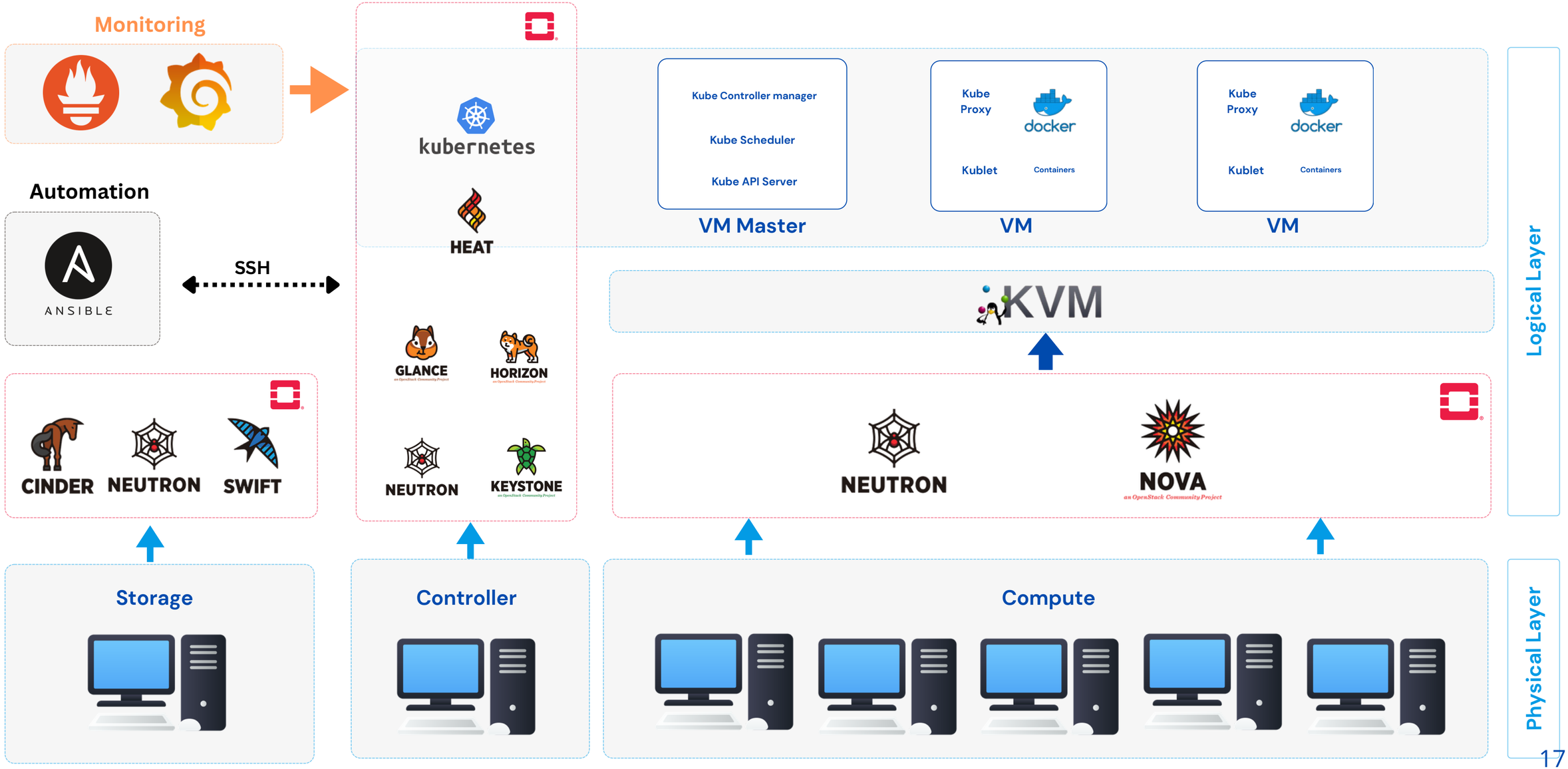


Object Storage

Logical architecture



Logical & Physical architecture



Thank you for your attention!

